

ActInf GuestStream 048.1 ~ Arthur Juliani & Adam Safron "Deep CANALs"

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hello and welcome everyone it's active inference guest stream number 48.1 on July 19 2023 we're here with Arthur Giuliani and Adam saffron also joined by Michael Edward Johnson in discussion we're going to have a presentation and then a conversation about deep canals a deep learning approach to refining the canalization theory of Psychopathology so thank you all for joining and looking forward to this presentation and discussion thanks Daniel I'm very happy to be here uh with admin Mike to talk about this work and to discuss more broadly um this General Direction that's emerged over the past few years to try to utilize some of the concepts from the active inference World um and computational modeling more generally to understand what's going on both in Psychopathology and in psychedelic therapy uh yeah so I'll give a brief a presentation and we'll move into a kind of more General discussion of this work it'll start uh with some kind of General considerations around psychedelic therapy some of the Neuroscience of psychedelics um and then move kind of more specifically into the model so there is this class of drugs the serotonergic Psychedelic drugs which include psilocybin DMT LSD and others and the past decade has seen a Resurgence of interest in psychedelic therapy psychedelic assisted Psychotherapy um where one or multiple of these drugs is used in conjunction with a kind of psycho therapeutic protocol um and there's been a lot of investment of money and time and clinical resources into utilizing these drugs for treatments depression anxiety disorders substance use disorders and others and there seems to be quite a lot of um positive signals suggesting that these kinds of therapeutic interventions can actually be uh quite effective um so there's a number of studies suggesting that psilocybin assisted Psychotherapy can be helpful for depression anxiety substance use disorder and others um and as this work advances um they're more and more larger clinical trials comparing it for example to uh the standard treatments with ssris with multiple studies suggesting that psilocybin assisted Psychotherapy is at least as effective as the Lexapro kind of didn't care all of this seems to suggest and the timelines are kind of pointing at this idea that psilocybin treatment

for major depressive disorder maybe only a couple of years away um from becoming uh approved as natural treatment in the United States which means that it's quite important for us to understand what's going on mechanistically in the brain so that these kinds of treatments can be best optimized best determines how to be used yeah or which kinds of mental issues people might be having and under what kinds of treatment protocols we might expect the most success I think this will also ultimately hopefully lead to the development of even better drugs even better treatment protocols um and much more personalized okay so some of the basics of how these drugs work on the brain for those who are unfamiliar right so they're these are the serous synergic psychedelics they are search energy because they act as agonists for a number of Serotonin receptors in the brain and the central nervous system more broadly um it's thought that most specifically uh these drugs like psilocybin work through agonizing the serotonin 5-HT_{2A} receptor and this has been demonstrated multiple times by using blockers antagonists of the 5-HT_{2A} receptor which seemed to largely remove the effect of psilocybin or other psychedelic drugs and what this 2A receptor does when it's agonized is it excites the neuron making it more likely to fire and this is kind of essential to some of the downstream effects so given that um it seems that this this 2A receptor that psychedelics are mainly targeting the question then become the school where in the brain or these two a receptors most widely expressed and research has focused on I would say three kind of key areas over the past decade or so the first being the thalamus which is a region involved in the gating of sensory information into the cortex there's a lot of expression in the cortex itself expression for example in the visual system which is perhaps not surprising given the prevalence of visual hallucinations and Technology use but therapeutically a lot of focus has been on the prefrontal cortex where there is also a dense expression of these receptors and then also more recent work has looked at the claustrum which is a small brain region responsible for coordinating the transition dynamics of cortical activation and this too is likely of essential importance to the Psychedelic effect so kind of moving up one level of abstraction to fmri studies there are a few kind of like key takeaways um which have I been identified from a few different uh fmri studies looking at um psychedelic use um kind of the main theme has largely been this idea of increased entropy or increased complexity of the cortical activation in the brain um so there are various measures of this um on the right here there's a figure of one of the common measures of complexity and looking at various different kinds of states that an individual might be in and how that relates to complexity right so you have individuals like vegetative States um or anesthetized anesthetized States having relatively low brain activity complexity

you have individuals in awake States like higher complexity and then individuals who who have ingested either psilocybin or LSD I'm showing the highest brain complexity now along with this uh rather than just this kind of like single unified score there's work looking particularly at the changes in functional connectivity between different brain regions or different brain networks um and the general Trend uh seems to be that you have decreased within Network functional connectivity and increase between Network functional connectivity right which this kind of can be interpreted as different uh brain networks which typically don't communicate as much with each other more likely to communicate with each other ones you the effects of psychedelics and the kind of like internal coherence of networks uh is reduced under people have maybe heard of this default mode Network right which is responsible for a lot of self-referential processing which is responsible which is typically activated for example when individuals are off task mind-wandering not like actively engaged um the default mode network has been kind of like consistently implicated in being having less of this kind of like within network connectivity less of this functional integration um and this has been hypothesized to be one of the kind of key mechanisms of the therapeutic effect of psychotics in addition to looking at kind of like brain network activity on a high level uh you can then also zoom in much more deeply and look at what's actually going on um at the neuron by neuron level or the neuron connectivity level here we see kind of very certain very consistent um story around this uh psychoplastogenic effect to this neuropathogenic effect where there are a number of different markers of increased plasticity in the brain so one that's very common is this increased uh brain derived neurotrophic Factor which we see hours after psychedelic use for a variety of different uh psychedelic drugs and this is then accompanied a few hours later by visible increases in the dendritic spine growth in cortical pyramidal neurons so this is both the number of dendritic spines and the complexity of the spines so you can see a very clear example of this here on the right where you have on the top the vehicle and then the bottom to DMT treated uh pyramidal neuron and the difference is very very Stark you don't even have to do any sort of um actual computational analysis for it to be clear that there's um quite significant dendritic growth from a single dosing and the changes to functional connectivity which I mentioned in the previous slide um tend to persist though in a less extreme form than what we see during the IQ effects so the question then becomes given all of this how can we attempt to tell a consistent computational story what's actually going on during both the acute facts the post-acute effects and in terms of the actual Downstream therapeutic outcomes so what's kind of been proposed um as uh the one of the leading models

of this is the relaxed belief under psychedelics model on the Rebus model of Carhartt Harrison and so this Builds on the kind of like active inference hierarchical prediction processing uh framework where we can understand the brain as the series of predictors stacked kind of like hierarchically along uh brain Networks um and predictions are being made by higher level Networks um that are attempting to uh predicts the errors of lower level and the process of basically doing this then produces this series of belief Landscapes at different levels of spatio temporal abstraction and these belief Landscapes of course can encode leaks with various levels of precision um and it's the modulation of this the Precision of these beliefs which is believed to take place under psychedelic use so here's kind of like a very high level uh version of this three of this model and how belief relaxation might actually happen right so given that you have this uh series of cortical networks which are representing beliefs of different spatial temporal scales we can um kind of think of the thalamus as being kind of like the entry point where sensory information is being gated into this hierarchical system and the prefrontal cortex is being kind of like one of the highest level networks right which is then responsible for um both representing the most abstract kinds of beliefs as well as being kind of the primary inhibiting force for all of the downstream beliefs in the network in the system right so what is uh this had the agonism of the 5ht_{2a} receptor supposed to be doing in this case um well it's going to be decreasing the prefrontal cortex's ability to make coherence um predictions and coherence representations of belief which will then disrupt its ability to inhibit lower levels of the cortical hierarchy at the same time by disrupt the this activation of the um salami or the destruction of Islamic gating right is going to result in additional sensory information entering the cortex that wouldn't underwise onto the cortex you have uh these kind of disruptions at these two levels which then results in uh beliefs being uh spontaneously represented in ways that they wouldn't normally be represented and the Rebus model at least is suggesting that this mainly is going to manifest itself as a kind of relaxation of these beliefs so here is a kind of like very simple diagram of what this might look like um so you have some landscape of beliefs here and you can kind of interpret the position along the uh the X and Y or I'm going to see x and z believe in a as being corresponding to some particular belief that might be represented by the network of this layer or at this level of representation and then the y-axis the height as being the actual Precision of that belief right um so here we have this kind of very deep part on the right part of this network or on this landscape and this would be a very strongly represented belief let's say what the acute effects that psychedelics have been understood to be doing right is to be flattening the Sunscape to be less strongly representing

who strongly represented beliefs to me making it easier for the um reversal of this landscape so to move from uh one attractive state to another attractive state which is meant to be understood to be kind of therapeutic in various ways and then the post-acute effect being some kind of return to normalcy but one that is not totally complete one that still incorporates some of these newly relaxed beliefs these newly explored beliefs building on this Rebus model um car and Harrison colleagues last year proposed this canal model of Psychopathology which basically kind of takes the intuition that through relaxation you know if the Revis model is correct if the relaxation of beliefs as a therapeutic outcome uh if you kind of take that way of thinking to its logical conclusion um you might then uh propose something like this which would say if relaxing overly strong beliefs is therapeutic then having overly strong beliefs in the first place is pathological in some kind of way um and they refer to this uh development to these overly strongly represented beliefs as catalyzed beliefs or and this problem generally is canalization or this process of developing them as analyzation and this is the canal model of Psych pathology so they kind of build on this idea uh this you know set of previous work in the literature suggesting that there may be some primary factor of Psychopathology and they suggest that this primary factor is canalization and in this work they discussed a number of different um psychopathologies which seem to quite clearly fit this um catalyzation type um characterization so it's very relatively straightforward I think to understand things like addiction depression anxiety disorders obsessive-compulsive disorders all of these involve in one way or another having some strongly held beliefs um typically negative negative beliefs uh in the case of like depression anxiety um obsessive compulsive body image disorder um or in the case of you know substance use disorder or some very strong beliefs about what kind of behaviors should be taken at any given time in any given context and then of course the effect of psychedelics is to undo canalization um and the better able any sort of therapeutic intervention is to undue canalization the more likely for a positive therapeutic outcome so I think this work has a lot of strengths and there's a kind of Elegance to it um in our Canal's work that we had been worked on we kind of identify a couple of limitations of this canal model and attempt to extend and refine the model in ways that I think ultimately make it more powerful while kind of keeping the the essential elegant Insight that canalization is kind of like intimately tied to like mythology so the first kind of limitation I think that we identify is that um in the canal model it's very heavily implied that all uh forms of canalization are pathological in one way or another um and as I mentioned just now uh this is clearly the case for a large Class A very common psychopathologies um it's when we start to look more

towards edge cases I think uh that these exceptions start to problematize this understanding right um so if we consider Psychopathology is such as bipolar borderline schizophrenia depersonalization disorder it becomes much more difficult to kind of easily classify these as simply problems of overly strong priors in many cases here we have problems of priors that are not so strong or priors that seem to drift over time in ways that are problematic for both the individual and uh the people that the individual are close to and if we think about psychedelics then as being this kind of Panacea to undo canalization and undo um Psychopathology we also run into this other issue which is that there are certain kinds of psychopathologies which are uh typically uh issues for use with psychedelics right so schizophrenia being a very clear one where there's evidence of psychedelic use being a trigger for the onset of psychosis and individuals who are predisposed to schizophrenia um there's also evidence of the development of depersonalization disorders with psychic with strong psychedelic use and this is not what we would expect if psychedelics were simply universally undoing uh or getting rid of psychopathologies there's also this idea which I think Adam is going to develop quite a bit later that the stability of beliefs are is typically correlated with positive mental health outcomes not negative ones and what is a stable belief if not one that is highly penalized in some kind of meaningful sense within within the foreign limitation is that if you think about an actual dynamical system one that uh is parametrized in some kind of meaningful way like a brain you know which is parametrized by synaptic connections and then instantiates uh neural activity patterns over time and we think about what kinds of optimization Landscapes are actually being induced by inside the system uh we find that there's not just one kind of catalyzation there's actually two kinds of catalyzation um and in our deep canals work we attempt to kind of take inspiration from Deep learning literature to tease apart these two kinds of penalization and understand them a bit more clearly and they are just to kind of give a a preview these two different things overfitting in plasticity loss um so overfitting is the problem uh you have your catalyzed in a particular kind of way and whatever sort of policy you've learned whatever sort of extent beliefs you have these don't generalize to some other context during context a you may be very well adapted you move to context B and whatever you've learned doesn't apply there then plasticity loss is a little bit different um elasticity loss has to do with one's not one's ability to generalize for other one's ability to adapt or to learn so imagine you're in context a you may or may not be well adapted to context a if you move to context B the question becomes are you able to adapt to contact speed or not not are you immediately already ready to deal with context B but given time in context B can you learn

and change and we use a very simple recurrent neural network model to kind of make sense of this and I think this is important because there can be different kinds of canalization on these different kinds of optimization Landscapes and these have unique implications for the treatment of psychopathology and I think uh even more importantly psychedelics work on these different landscapes in different ways and so understanding how they work uh is essential to developing better drugs better treatment protocols and more personalized treatment Okay so what are these two optimization Landscapes how can we make sense of them this is kind of like the big overview figure of the deep canals work so we kind of start with this intuition of our current neural network right and in a recurrent neural network there are these kind of twos there are these two sets of parameters so you have at any given time you have the actual weights of the network and then you have the hidden activation of the network right and so the weights of the network are data and the hidden activation is H and there's some input into the network at any given time step u_m and then the output of the network is a function of both the input and the hidden State and the hidden state is updated at every time the hidden activation pattern great and this gives us essentially these two different Landscapes uh one what we refer to as the inference landscape or the type a landscape and this is equivalent to the neural activation pattern of a brain at any time right so what gets read off from an fmri what gets read off from the EEG is going to correspond to this activity landscape but uh kind of like underlying or supporting this activity landscape is this other more fundamental landscape which is the actual optimization landscape that is induced by the synaptic weights themselves and this is what is the result of uh right like synaptic connectivity in the brain between all the sets of neurons in the brain this is what is changing when we have various forms of like heavy and plasticity heavy and learning and this is what kind of like instantiates the possible set of neural activity patterns right so this is kind of like a way we can start to understand this which is when we think about belief Landscapes right as I mentioned before you know here's a very simple diagram where we have a two-dimensional belief landscape and let's say there's uh you know someone who's depressed or potentially depressed and they might have a negative self-image you know I'm a terrible person I I can't do anything right something like this this would be encoded as a particular position in this belief landscape um you know where the neural activity pattern looks like this this belief is what gets expressed right and there's another you know set of neural activity patterns which would correspond to a positive self-image and positive self belief you know I'm great I'm wonderful or even just I'm okay I'm totally acceptable as a person um and this would be a different set of patterns so uh

based on the synaptic the actual connectivity structure of the brain right uh these two patterns May or be maybe more or less likely so here these are being represented by these uh two um sets uh neural activation Landscapes which are the result of two different positions in the synaptic weight landscape space and in this top space um the one that's in the middle here uh that then is represented here at the top of uh part C of the figure we see that there's very high Precision assigned to or very high likelihood assigned to the negative self-image and very low likelihood here assigned to the positive self-image but in some other part of the uh synaptic weight landscape a very different kind of neural activity landscape might be instantiated um and this one kind of has the inverse properties where negative self-image is unlikely and positive self-image is much more likely um so we can kind of then understand positions in the neural activity landscape is kind of corresponding to specific phenomenological experiential instantiations of certain kinds of beliefs right so like the actual experience of uh holding a negative supplement or holding a positive self-image and we can understand positions in the synaptic weight landscape as being something more General so in the case of depression um there might be a specific set of synaptic weight connectivity which would correspond to what you might call you know a depressed individual someone who might qualify as having a major depressive disorder and another set of synoptic weights connectivity pattern that wouldn't qualify as that way okay so before I get into this psychedelic uh oh psychedelics relate to this um I think Adam is going to talk a little bit about um some work he's done extending the Rebus model hello so yeah I've been uh thinking about this a while and then along the way I met Arthur who also has been thinking about this a while but in different ways and so um uh like Arthur's background isn't um basically Consciousness science and a very uh peculiar uh field of computational Consciousness science in terms of actual models of what it would be to have a conscious system uh even potentially an artificially conscious system and so um I'm coming from more free energy principle and active inference my background and Arthur's coming from more mainstream machine learning and so we we're coming together to explore uh this beast from different perspectives and been finding a lot of uh I think very interesting overlaps and so uh in terms of Albus the basic idea is to take Rebus um and expand on it we I love Rebus it's a great framework um and I'm I've been trying to make it I guess more flexible in terms of being able to handle like a wider range of assumptions so not just talking about relaxed beliefs but also potentially strengthen beliefs in different combinations uh this was always part of the Rebus framework and that you can have all sorts of both direct and indirect effects when you're dealing with a large predictive hierarchy or any you know complex system

you change something in one way but then you'll get an effect elsewhere so if you have this hierarchy of predictions for instance you can potentially relax beliefs at higher levels and that can indirectly cause strength and beliefs at lower levels so you can both get relaxed and strengthen beliefs in psychedelics in all sorts of different combinations and so with Albus the idea is um so I've been calling uh as opposed to a belief relaxation when you get this sort of um basically a bottom right here you might see it so this red would be basically uh a something like a synchronized complex of neural activity um and potentially you've often involving the thalamus which would be thought to encode your predictions or your top-down stream of priors which then are suppressing the ascending stream prediction errors um and so under Rebus you're seeing here uh relaxed predictions a smaller uh complex of synchronized activity and the gates are down just like Arthur described more things are coming in Rebus will sometimes be described in terms of uh so when we're talking about Rebus this idea of relaxing beliefs we're thinking at the um functional or computational level so like David Myers and analyzing the mind you have these three I suppose uh supervenient levels out there compatible and synergistic so computational functional um like abstractly what is the system doing uh something like the pseudocode of the system and then implementational mechanistically how's realized and then in the middle some sort of algorithm algorithmic intermediate abstraction layer how are these computations realized via these mechanisms some bridging principle description and so when it comes to Rebus um you can adopt it in different ways and relate to it differently so you can relate to it at the functional level um or you can relate to it at the algorithmic or implementational levels um in terms of where you're taking the account so for instance uh you might think something like Rebus effects is a very valuable model of something like what they call Quantum change with psychedelics but you don't necessarily believe in predictive processing but you still might be you could say like a Revis type theorist or you might be thinking in terms of like Rebus like in a more narrow sense of thinking of uh predictive processing information flows um but then when you get to like the overall person and integrated organism and the agent uh you might have different interpretations there um so with the work with Albus basically the idea would be to take predictive processing and to and models of psychedelic action and suggest that sometimes um not only uh do we have these relaxational belief Landscapes but we might have the opposite of strengthening and so specifically if you're activating these five HD 2A neurons which would be at the deeper levels of the cortical sheet and also the deeper levels of the cortical hierarchy more inwards and higher up and these will be the Unice Loop of the thalamus and thereby be

able to form large synchronous integrated complexes um but then when you excite them too much they don't wait for each other to sync up they fire or whenever they can how to sink and this relaxes a belief landscape however it's not clear this should happen the whole way through for any amount of increasing the gain on these neurons So in theory just increasing the gain a little bit unless uh that might help them to sync up that might strengthen the belief whether it's a situationally specific suggestion of you in this context or maybe something old coming to the surface and uh if you will manifesting um so Albus basically in a nutshell would be saying um to take Rebus and keep going with it and look for different combinations of both strengthened both directly and indirectly strengthened and relaxed that's basically it and so I'm not going to go into too many more details Arthur and I just submitted this uh for publication uh wish us luck hopefully the reviewers will be useful um maybe even kind but preferably useful and um yeah I'll kick it back to Arthur but before I do um like this work uh I think it's immensely important that we get these details right these are um extremely powerful medicines that could revolutionize things or we could have a backlash if we do it wrong if we're careless if we um don't see the risks but also if we're overly beholden to like a single way of viewing things prematurely we might also miss opportunities to help potentially enormous opportunities for instance like you might not think of helping people um like look for instance micro dosing for cognitive decline you might not have thought of that um and that might be a really big deal that you do think of them so uh I guess before going back to Arthur one more thing would be just in terms of uh at least for me like experientially like the way I sometimes think of it as like um with relation to well-being and uh suffering both things matter like the strength and the strength of your positive beliefs and the strength of your negative beliefs and sometimes the negative beliefs have positive aspects and vice versa so it's it's both like you might relax your beliefs and like after he's like a Harry Potter metaphor like be someone like out of it so much out of the grip of the Dementors that usually uh that under some circumstances might be pursuing you that you have room to be different to become different uh to think different things under this different state um but additionally and not um mutually exclusively not necessarily in compatibly maybe synergistically the strength of what beliefs you're able to have in that moment and maybe draw upon from the past could be deeply important uh your values the kinds of beliefs that you call upon in times of need and that give you hope um maybe those are just originally suggested to you right there in the session but maybe there's something else another belief in that strengthening it could be a good thing and even without a clinical context like microdosers for instance it's not clear they're exactly they're trying to relax

their beliefs like the range of use cases the way we want to relate them how we want to be in the world it seems complicated to me so I will get back to Arthur yeah thanks so much Adam so with that context with all this uh I want to talk a little bit about how we might understand uh at a very high level what psychedelics are doing to these two uh Elite landscapes so before I showed you very quickly I guess I can go back to it um in the Rebus model you know there's only one type of optimization landscape uh and so there's only one type of uh canalization and what is you know supposed to be happening is a relaxation of beliefs um which is strong during the acute effect and persists in the post-acute effect so with uh Elvis though we might be able to think about all of this a little bit differently um we can in during the acute effects of psychedelics interpret what's happening to the type a landscape as being less of the kind of universal relaxation and more of a kind of destabilization where under certain circumstances beliefs are being strengthened other beliefs are being relaxed this is uh changing quite rapidly over the course of time so you know a certain belief may be transiently strengthened a few minutes later it may no longer be supported or supportable and this actually functionally you know enables a kind of exploration of all the POS of a wide space of belief instantiations during the acute effects of the drug the uh as far as the inference landscape is concerned you know post-acute effects as Adam was saying it could be the fact that certain positive beliefs are explored and can be strengthened um other negative beliefs might be weakened this is kind of like the ideal psychotherapeutic outcome it may be you know on average if it's something like a relaxation a general relaxation of beliefs is uh what you might see but the way that you get there is likely going to be through a kind of much more complicated process I think what's that interesting and related to also to what Adam said about uh thinking much more broadly about how psychedelics might be used in terms of neurocognitive disorders in addition to kind of like traditional psychopathologies um is to think about what's happening to the the learning landscape the kind of like underlying synaptic weight landscape and what do we understand is happening here kind of quite clearly is there is uh neurogenesis psychoplastic facts um there's dendritic growth and what is this enabling well this is enabling uh greater connectivity between within and in between networks and so kind of from a functional uh learning perspective learning theory perspective what this means uh is that gradient information can flow more easily through these networks um and what this means is that it's easier to get from one part of the network to the other part of the network there are fewer of these kind of optimization uh local um Minima there are fewer of these kind of like tricky saddle points um and what this means kind of like functionally is that this plasticity loss that I discussed earlier

is going to be less of an issue and here it seems to be kind of like regardless of the acute uh the actual phenomenological effects uh you know there are very strong Downstream effects on uh this structural connectivity and I think um you know there's some interesting work suggesting that there may actually be independent mechanisms um that are affecting these two Landscapes so there's a reason we're suggesting for example that uh psychedelics as well as ketamine is able to kind of like particularly bind to receptors in the brain that produce that are results in the kind of release of the brain derived neurotropic factor and then the downstream changes in synaptic weighting and that this may be uh potentially totally independent of the 5hc2a agonism right and this has very interesting implications for future drug design and development okay so I'm going to before my next slide turn it over to Adam once more uh to briefly talk a bit about personality theory and the relationship between uh psychedelics and personality hello again so um so for the past few years I've been doing some work with a personality researcher Colin De Young on intersections between the way agency is modeled in the free energy principle and active inference framework and the way that persons are considered in different situations and personality science and looking for points of intersection and so um you know it's personally personality is actually an extremely deep concept and if you think of it it's um I guess we call it like the normal form of a system in terms of it's uh what is the way you can describe the most of the system um with like the minimal message length the um account for the most with the least and what we're trying to account for is the states that people tend to occupy people aren't always happy people aren't always sad people aren't always you know friendly they're not always angry it's all sorts of things that are there there's a range that people are encountering but there's they tend to occupy some states more than others and you might call that the person their personality and you might call that the person many systems would have their own Personalities in terms of trying to describe uh something with their essence and so uh they're attracting States and so with uh Colin the Young's work um he interprets personality and these um major domains of Personality that people have discovered uh a reliable Factor structure ways you can Funk the variance uh the big five as a trade hierarchy but where these different traits are interpreted as cybernetic control parameters uh specifically cybernetics in terms of the study of goal seeking systems governed by various forms of feedback that different stages of uh the process of pursuing goals uh evaluating outcomes updating um that and you understanding that basically the personality as different modes of being in the world for cybernetic systems interpreting them as parameters for goal Pursuit because we are goal seeking systems trying to pursue the goals

we value I guess moving to this the and with relation we're talking about now both in terms of Personality formation reformation and the structure of Personality the piece so okay to get to the P Factor uh within personality theory this big five traits the set of big five traits there's a uh richer or more detailed uh Factor structure um a hierarchical uh structure for personality that's been discovered um so above the big five neuroticism agreeableness conscientiousness you'll have a metatrase of their shared variances that's observed called stability it's also been called Alpha by think digmin um and then above the traits of extroversion and openness intellect you have this metatrade of plasticity and um are called beta and the shared variance of these two so what's interesting is that within uh this field of um within high top which is this effort to try to characterize Psychopathology using the same kind of data-driven methods and personality science a principled account of the structure of Psychopathology the way that personality has a structure you're trying to account for this there seems to be a common factor across many forms of Psychopathology that's referred to as the P factor and it tends to be the inverse of stability it's high neuroticism uh low agreeableness and low conscientiousness and this is actually uh uh what's referred to in the article um as um this P Factor thinking of it as a lack of flexibility you could make a case for it but more straightforwardly it seems to be a lack of stability for something to be a lack of flexibility you would maybe be thinking of something more like uh plasticity like someone having low plasticity but empirically that's not what's observed and so empirically that's not what's observed that's a very important detail we should stop and take note of so you might say that um rather than emphasizing the first principle component of Psychopathology it's possible that this canal models is actually focusing on the second Factor um it's not you know it's not a contest it's still enormously a big factor like the work of Stephen Hayes and others showing like a lack of flexibility is trans diagnostic but just empirically speaking the first principle component of Psychopathology might be more closely related to a lack of stability rather than insufficient plasticity um well I guess one more thing to think of in terms of stability and plasticity like as I said it's not like a it's not exactly a contest uh they're um they're both important both need to be considered and they're deeply interrelated so Colin describes them as being in a state of something like Dynamic tension like if you're insufficiently uh plastic and then flexible you're not going to stay stable for long in a changing world uh but if you don't have a stable base to move from uh it's going to be hard to be plastic and reconfigure yourself in a sustainable way either however you can also see someone being over um overly rigid and excessive stability in a way that undermines their plasticity or overly flexible in a way that basically is destabilizing so a dynamic tension but

involving synergies also and I think in terms of the structure of personality and well-being this is what we need to keep in mind that both are needed stability and plasticity and this is uh also just within um it's actually what this was inspired I think of in Collins thinking to consider these metatrades this way uh from machine learning I think actually reading grossberg and stability plasticity trade-offs and that's why he interpreted these meta traits in those terms uh so I guess one more thing is uh there's a petersonian backstory here they don't know if I want to get into but we can discuss it later if there's interest so yeah back to Arthur that was great thanks Adam yeah very very helpful context which should hopefully make uh this slide here much more clear to everyone so given these kind of like two meta traits uh stability and plasticity which as Adam you know beautifully described her in this kind of like Dynamic tension with one another we can kind of go back to these two optimization Landscapes and these machine learning Concepts that I discussed earlier and kind of try to understand how these things might all relate and ultimately you know relate to so I had hinted before that they're were these kind of two types of canalization we might say one being overfitting and another being plasticity loss and we can interpret these as kind of like being extremes uh that kind of like analyzed end of uh how these two optimization Landscapes might manifest so in the type a in-print landscape this looks like overfitting on one end heavy canalization and underfitting on the other end in machine learning context right so overfitting being uh having very strong rigid beliefs that are in a particular environmental context adaptive that are very unlikely to be adopted in any other context underfitting being having a set of beliefs which are not adapted to the current context and also unlikely to be adapted to any other context awesome as far as stability is concerned I I think we can understand stability this metatrader is kind of representing the balance of these two things right you have high stability when you are an either um overfits to a particular contact store underfit um so this means you're well fit to this context that you're right now and you're also likely the wealth that's other future context um I think this is also in complexity science sometimes described as meta stability right um so you're in a stable Point that'll also likely be able to take you to other stable points in the future and this kind of like ability to stay stable over time is then uh stability metatrade we uh can then look at the type B the learning landscape where the kind of extreme version of canalization is this plasticity loss where it's very difficult to adapt to new contacts the inverse of this is then catastrophic forgetting where the problem is not one of adapted to new context the problem is one of meaningfully retaining adaptive strategies and beliefs from previous contexts and so obviously you know neither of these are desirable what is desirable is some

kind of balance between these two right where one is able to adapt just enough to new contacts without being uh without forgetting previously learned things but here the relationship uh to the metatrade of plasticity is a bit different than stability right so as it is being represented on this crop um you know the more plastic you are basically the less analyzed you are in this space the more easy it is to move around um but actually extreme versions of this end up manifesting is catastrophic forgetting police in the machine learning context um and I think this in particular has relevance for psych pathology and relevance for this idea maybe complicates this idea right that uh psychedelic therapy is uh in and of itself likely to always lead to positive outcomes okay and we can attempt to understand this by uh making an effort at trying to map or understand um various forms of Psychopathology in the context of these two different learning Landscapes um and so before I get into this too much I think it's important to say that um this is very much a kind of like preliminary and provisional attempt at making sense of these things um I hope that this is kind of the start of a a much more fleshed dialogue looking at how we can kind of understand Psychopathology with respect to machine learning Concepts um but I think this at least gives like a good starting point for maybe trying to understand where and why um psychedelic therapy is useful isn't useful which is a bit more nuanced than the simple idea of like Psychopathology means canalization and psychedelic therapy means reducing canalization right okay so let's uh try to make sense of this a little bit though for the kind of classic set of Psychopathology is for which there has been a lot of great research done with psychedelic therapy and a lot of clear benefit has been seen um I think you we can kind of class these within this upper left-hand corner here where these are disorders of both overfitting which means in the moment over the strong beliefs get represented you know these are strong attractor points in the neural activation uh plasticity Loft in the type B landscape which means that not only is a certain kind of Dynamics instantiated right which make for example in the case of the major depressive disorder extremely negative self-evaluations very common but it's the case that even given the correct changes in environmental context even in the you know even given in the case of you know psychedelic system therapy a lot of the work with major depressive disorders being done with treatment resistant depression right and what is treatment resistant depression but uh a kind of a set of beliefs a set of neural activity patterns which are resistant to adaptation to new environmental contexts being you know whatever sort of psychotherapy or previous drug was used so this is a high plasticity loss however fitting right I think we can then uh and these are you know kind of unfortunately you know for psychedelic therapy and I think maybe for

psychiatry in general these are all very very common and very very costly uh psychopathologies right so if it is the case in psychedelic therapy can treat these effectively that's fantastic but I think we can maybe look at a more complicated you know if we start to expand to other psychopathologies um this story becomes a little bit more complicated right um so let's look we can look at the bottom left-hand corner here where we have cases of in the inference landscape still overfitting but um in the learning landscape something else happening a form of catastrophic forgetting or form of kind of drift of the belief landscape that's being instantiated here right so I think two kind of Representative um pathologies here are typically referred to as some form of bipolar disorder or some form of borderline personality disorder right so in both of these cases in the moment to moment uh neural activity as it unfolds there are very strong sets of beliefs being instantiated at any given time and these are very difficult to change right um but the key differentiating Point here being that over longer time scales over the time scales of days or weeks um the kinds of beliefs that are instantiated change quite a lot right so someone who's bipolar if they're in a manic episode or in a depressive episode the nature of the belief Landscaping instantiated is very different um for borderline individuals from a daily or weekly basis the nature of one's beliefs about the people that were close to them can change very rapidly um very extremely very in certain cases and so here the benefit of psychedelic therapy uh is a bit more mixed where there is likely the case that to the extent to which the uh kind of overt type A beliefs can be updated and changed to be more positive be hopefully more relaxed um that's positive but to the extent to which plasticity may be being increased when it's not actually needed when there's not actually like a meaningful therapeutic benefit to that um it's a bit questionable I think you know there's clear borderline or rather bipolar for example is often individuals with bipolar are often excluded from psychedelic files under the assumption for example that psychedelics may trigger a manic episode or may trigger psychosis um and I think that is a valid concern um so on kind of that topic we can then this bottom right hand corner would be cases of both underfooting and catastrophic forgetting with I think schizophrenia being kind of like the most clear manifestation of this and here um you know there's very little work currently looking at trying to treat uh schizophrenia with psychedelic therapy I think for pretty good reasons though I would say to the extent to which there are secondary uh manifestations secondary issues like depression anxiety that people with schizophrenia might have you know these actually may be treatable um with some form of psychedelic therapy and then lastly this upper right hand corner here uh would correspond to underfitting in the type a landscape along with

plasticity laws so I think a good example of this is something like ADHD um where there are not very uh particular strong beliefs that are being instantiated at any given time but it's difficult to change these side of beliefs nonetheless um I think here there's a case uh where it's not totally clear on how the benefit will actually play out in both of these cases I you know I've heard interesting work around like micro dosing maybe of issue interest I think yeah more research has to be done to be seen right and then what would micro dosing be doing right micro dosing would be like very slowly uh kind of like decreasing plasticity loss just a little bit um potentially also decreasing underfitting right as Adam had kind of suggested in the office model and here we have just kind of like a brief overview of more or less everything I've described about how these two Landscapes are different um I think most of these things I've said before pretty clearly right so the type A landscape it corresponds to the neural activity pattern this is the hidden state of the RNN over canalization is going to be stereotyped mental circuits very strongly held beliefs under canalization is going to be these inconsistently deployed circuits it looks like uh overfitting in the canalized case underfitting in the non-catalyzed space going to correspond to stability to metatrane um I hadn't discussed this before but um I think there's a relationship between cognitive flexibility and this inference landscape psychedelics are mainly going to have faculty uh work at the acute level care and this is by directly exciting the neurons or inducing forms of how to be in plasticity and the outcome is going to be a mixture of relaxation and strengthening of the kind that Adam described in the the Albus case the type B landscape then is working on the synaptic weights and these are the actual Network parameters in our RNN model over canalization is this inability to learn or adapt over time under canalization is the inability to retain information right and those correspond to plasticity loss and catastrophic forgetting this also is kind of the traditional plasticity not a trait um which I think corresponds to this more uh this longer term constructive psychological flexibility um the effects are going to mainly be mainly to post acutely with psychedelics and these are the more Downstream metaplastic effects um the psychoplastic effects um and I think on the whole the outcome the effect here is a relaxation of catalyzation and uh yeah these are just the references so those those are the slides I have that's kind of the overview of the deep canals model um I think now I'll turn it over to Mike who um who's going to share some thoughts both about this work and some of his own work and kind of this uh approach to understanding psychedelics more broadly uh great uh yeah thank you for the that overview that was great um so I guess uh I'm just in general very very optimistic about this corner of uh of Neuroscience that uh there are these very intuitive

models uh that are very sort of clinically significant and and I also think you know there there is a lot of responsibility to sort of get things right here uh as well because as I remembered so uh yeah I I have just a few comments um and uh for the for the viewers uh I I'm not speaking uh for for Arthur and Adam um I'm just grateful for uh the invitation to join the discussion uh so I wanted to to know uh quickly three research teams dealing with the implementation of this class of model uh so just broadly kind of pointing at uh both deep canals and uh andrewbis uh and uh and Albus um so um in 2018 I wrote neural annealing and that was sort of uh developing the canalization frame within the context of energy buildup in the eigen modes of a um so you have the the nice annealing metaphor uh you have a temperature parameter and entropic disintegration uh uh Robin character said a a beautiful paper on that um uh potential brain and uh then you you sort of have a post uh heating result uh where the microstructure of your system uh is sort of a result of the uh sort of how hot and how long the the heating was so it's it's an alternative take on belief relaxation and I do think that like there are some pluses in terms of sort of explicitly making the annealing metaphor um and uh uh I think that's uh like two things that I'd point out uh one top-down models or top-down predictive models as energy sinks um and eigen modes as the places where energy can build up for almost sort of hide from top-down models um so you get this uh this phenomenon of semantically neutral energy or semantically illegible energy you can say uh which is sort of kind of how uh perhaps uh psychedelics music and rotation all sort of allow energy but um they're the energy that they contribute to the system can evade the top-down models um and uh yeah just to note one item that I'm looking at now is what is deep canals has the inference landscape versus learning landscape distinction and how would one sort of build that distinction within the neural annealing framework and uh this one one metaphor that I'm I'm considering is the self as sort of a magnet uh to sort of uh apply a constraint a uniform focusing constraint on the system as it uh as it's cooling um the second uh item that I I want to mention that so I have this piece autism as a disorder of dimensionality and uh this brings in the concept of network density uh that it may very well be that uh people on the Spectrum have literally more neurons and or more synapses and there's this great Michelangelo Michelangelo quote I saw the angel in the stone and I set him free um and sort of the thicker networks you have the more Stone you have to work with but you actually have to do the work um so uh you you sort of have to have to do sort of well normally you sort of inherit these highly pre-trained optimized circuits almost like Asics application specific integrated circuit um and then uh this is kind of more uh an untrained Network and you have to turn it yourself um and uh within the Deep canals

framework this would show up as both underfitting and a higher capacity for plasticity so I think that's uh that's gonna be a an interesting thread finally uh so I just posted that last week principles of vessel computation and as an implementation level account of active inference and this is a big rabbit hole um I I don't think we have time to to go down today uh but I just comment that I really like the way deep canals explicitly splits out the learning landscape and the inference landscape that uh I think that making this distinction is increasingly going to make sense uh back to back Arthur yeah yeah awesome well thank you all for the presentations Arthur and Adam feel free to to pick up where Michael left off or I can bring in some questions from the chat or from what I've written down all right just uh yeah just a quick comment um I do think that uh in in the Deep canals framework it's going to be interesting to sort of look into uh metabolic demands uh and like sort of uh like metabolic disorders are are pretty hot in brain science right now and I do think that uh it'll be interesting to to look at sort of what the metabolic profile is in terms of each of the four items of a quadrant if one quadrant or another quadrant is particularly associated with uh um more metabolic demands in the system Arthur yeah yeah that's fascinating I was I'm actually I'd love to kind of hear a little bit more about this comment you made about the cell as a magnet that's applying these constraints oh yeah so there's a the like psychedelics the the effects are not or sort of the you sort of get pulled back into position after some time and uh you know I can't say that this is a full full Theory uh at this point but um in the Neuroscience meditation 2018 I talked about um the self as kind of the the god of the gaps in terms of aligning different scales of uh like you you have circuits which kind of pull in different ways and you have the mesoscale systems Which pull in different ways and you have the the overall brainwise systems that pull in different ways and kind of the self is sort of the thing that I'll try is to make everything work together and an alternate frame would be sort of this self is just kind of this slow gravity toward disalignment uh yeah um I don't have anything particularly uh more to share that's it yeah no that's interesting well to combine the heating up and the magnetism different metals and connections there to Alchemy and everything different Metals when heated up and cooled down in specific ways change their magnetic profile so of course this is all just concordances but they're very interesting and then also to the um the metabolic angle that reminded me of the um the trilogy by Dale pendell with uh pharmacist and and the other two books where treated alongside traditional psychoactives we could say like caffeine and nicotine and some of the other substances that were mentioned here also there's tea sugar and these are treated as psychedelic or at least as uh in the same Pantheon of neurocognitive modifying in the sense that the

Destinies of individuals and of Empires ride on the uh sense making and decision making influence of sugar for example so it definitely broadens our perspective to include the the so-called classical molecules like LSD but also maybe we could think more broadly about antioxidants or about different lipids or about different metabolic connections Michael Pollan's uh recent book *This is Your Mind on Plants* I think like it kind of has that what you're saying in mind because like it includes caffeine um not like introduced as a psychedelic but just disgust among uh also psychedelics I think their memory focuses on masculine but um just it is a different way of being in the world uh potentially radically different like if caffeine is in your life where it is not it's a different life world and different extended niche um I'll go to a question in the live chat so anyone can give a thought so Matt asks is there any way how to approach the individual reactions to psychedelics in terms of positive or negative results of treatment using the ideas of deep canals so how can we approach predictions and explanations at the individual scale with what you're sharing today one thing for wow this is kind of like a meta comment but like if we're actually going to do um computational Psychiatry and we actually are going to and precision medicine so look at the goal being to create models of sufficient detail that they could actually guide our practice and guide our practice in a way where we know for an individual what they might need in any given context so Precision medicine with respect to like standard antidepressants um you know people sometimes will go for a long time before they find ones that work and sometimes things are made worse and they might deal with antidepressant withdrawal as they're looking for the right one and if they're in a state of acute distress that could be um a hard lot and so like if you can help to speed up that process of finding what works uh the ability to and what might hurt also to avoid that I mean the ability to help people's immense and so for that it's um like part of this so the meta comment is like um I have to say I'm not convinced that an active inference uh we're doing the job I think we're not acknowledging our uncertainty and I think the models are underspecified with respect to the complexity of the effects we would expect on different scales and in characterizing the systems and we need to be more patient and modest but what we don't know before we go on rushing in headlong into telling the rest of Psychiatry how they should comport themselves and what models they should use uh based on psychedelic science and plausible models hopeful models but very early days and with the reality being complex in the ways we're seeing here like this was not an easy set of lectures you know or a set of talks like what we're going through this is like and yeah if we're actually going to be able to handle these powerful medicines responsibly we got to go into the weeds and so

anyway so the one more meta comment was like I think I saw like someone say oh how can something like help everything you're like yeah it's almost I don't know if this is right it's almost like a violation of like no free lunch theorems or something it's like how could something just be good across the board without you know something is off you've calculated wrong somewhere um you had to pay the price either in like some trade-off that wasn't acknowledged or um but that being said there are some things which are generally good for a lot of things like being more flexible or like if if you can get something like meta stability like like I said there's almost like I'm I'm I think I'm a church member of the the or I'm a faithful member of the Church of criticality uh I think Mark Miller might have uh indoctrinated me with like Optimal grips but basically like the ability to have like an optimal grip on existence in your soul not too tight not too loose and being able to move around and be like flexible in itinerant like something kind of like mindfulness is just like Sriracha for the soul it's good for everything almost you just need to put on this put on that makes it better I I I I buy that there might be some things that are good across the board but I wouldn't call though uh psychedelics necessarily um putting you in this like state of optimal criticality or anything it's like in general like the rhetoric I'm uncomfortable with of saying psychedelics do X I would prefer to say like um which psychedel like psychedelics tend to do X but just a little more qualifier like not weasel words but just like a little bit more like uh this psychedelic might do X in this context with this dose and this sentence setting but that being said there do also those seem to be across the board um broads um sweeping not super but overarching statements that we can make that Express a lot with a little and so like I don't want to be so much of like a splitter and like a a prickly person finding all the details that like these powerful things we can basically give to people to guide like our research I got I don't want to leave that on table either so yeah so I guess um that's a bunch of metal level commentary that did not address the thing so I did that too so that's very helpful I think to have all that context um with respect to the question itself I would um maybe offer this which might be helpful which is you know I think you mentioned it Adam and it's very frequently mentioned this idea of satin setting right as being like heavily predictive of the quality of the acute psychedelic experience and often also to some extent the quality of the post-acute experience um and I think we can understand this in the Deep canals model quite quite straightforwardly so I'll just quickly pull back up one of these slides yeah right so what is satin setting so um set is mindset which would basically be some combination of the current uh your current position in the synaptic weight landscape and the current state of your neural activation landscape right so your place in both Landscapes is

your mindset and uh the setting is the what is generating this input right what is generating the X into the network and that's you know some complicated function of the environment um and then we can understand right the evolution of movement through the inference landscape during the acute effects of the drug as then kind of like vary directly and straightforwardly being a function of these two things um because both of these these things are are basically what's affecting how this landscape evolves over time and how you move through the landscape um and so you know with this in mind I guess we can then start to characterize and think about like okay well what kinds of uh you know we can maybe be a little bit more mechanistic about like how does input from the environment the setting actually given you know that we relatively fix the actual synaptic one of the way landscape how is that changing the Dynamics over time or the other way around right like how our uh is fixing the input the environmental context relatively stable um and looking at individuals who are at different fixed points at different points of time I'm going to change the Dynamics but yeah the event right and I think given that this is a highly stochastic process given that these dynamics of Albus are are at play right where like certain beliefs are being strengthened other beliefs are being relaxed I think at least having this hold on okay well set and setting is really important and it's important because like these are explicitly conditioning the way in which uh both the landscape the inference landscape is shaped and how you move around it and how it's being you know dynamically perturbed um yeah hopefully that answers a little bit the question nice uh I just want to jump in there just thinking about um biomarkers I mean if if we're looking for something that's of high clinical specificity uh we want to sort of be able to measure what's what's going on uh at a very careful in a very careful way so like first of all what is temperature parameter uh you know we we talk about okay psychedelics increase the temperature parameters but um like how high and do do different substances and dosages uh sort of differ in interesting ways probably yes uh and then also sort of measuring temperature parameter measuring the temperature parameter in different parts of the body so you might have a different uh temperature like not the real temperature but uh kind of computational like entropic concentration uh in different parts of your body um and also um the plasticity status in different networks throughout your body so I guess I I see a big Frontier here uh as thinking some doing some careful thinking around um biomarkers and proxies and inferring these these sort of things that we we are talking about as core and I think they are core and we should be able to find a good uh good marks yeah that's totally correct and yeah I just want to like repeat it for my own occasion right which is it like yeah these two things which I think were maybe

one was is I think already a core part of the deep Penance work right which is that I was showing these single images of these toy optimization Landscapes but of course the human being the nervous system right uh has multiple many of these Landscapes hierarchically organized right um and they influence each other in a complicated way the kind of like yeah the project for it is to understand where these Landscapes are what they look like and how we can find biomarkers to them yeah and again you know if he's two kinds of characterizations are true um this overfitting underfitting plasticity loss and uh catastrophic forgetting then it would it's a very meaningful project to be able to you know quantify that objectively somehow an individual and that's when you can start to get to this point that Adam was talking about like Precision medicine and understanding like okay well actually it turns out right like these networks are catalyzed in this kind of way and because of that we can say with much more certainty that this kind of you know psychedelic intervention is going to likely be helpful I'm wondering I don't know if this like potentially brings in some of the work with like um uh vascular and neurovisceral uh neurovascular and visceral integration that you're doing but like so there's like a sense in like the free energy principle where the overall phenotype and an extended phenotype in terms of its cut its constructing and coupling with and constructing its Eco Niche there's um it's success and manifesting this um it's there's one for energy function for this overall effort as a kind of prediction that's enacted through the Dynamics but you could also like break it down in terms of like subsystems with like local gradients and like local auto poetic processes that are in like they're brought together into an overall system hopefully with synergies um but and they have to be to be in an organism and an agent we call this like I think to the extent it has thickness it's coming together but I guess so you could I guess think of maybe like a energy landscape you know something like uh let's just say like right now in this moment like my uh explicitly felt beliefs but my my body also might have a different opinion different parts of it and it in terms of like what it's implicitly optimizing it's revealed preference from its in action that may or may not correspond to how I feel about it I might not even it might be difference of opinion I might even be tuning in I might be ignoring it because I don't want to and so it's kind of like the score kept by the body might have its own um uh be describable as its own like information geometry which is being tracked to degrees and is in different degrees of uh alignment with um different ones at different levels of a control hierarchy so plausible right yeah absolutely I I think so um yeah uh and like I I think of I think the body is sort of um uh a parliament of different uh networks and uh you know alignment is the scarce resource coordination is always a scarcity resource and uh like how

does the body keep alignment and when does it go wrong too and I guess I see this as sort of um I mean harmonically mediated each part of the body has some choice of what else in the body to listen to and depending on the particular sort of location in the information topology uh different parts of the body have different carrots and sticks by which to try to control other parts of the body uh so if your stomach is very unhappy uh it's going to start rambling and then it's uh it'll let the body know okay I need some food like I really seriously need some food um and then yeah the rest of the body they can sort of turn away from it uh and then it'll just get louder and then you have to turn it away from it more a little bit louder and that's a of course a negative Dynamic if it gets What It Wants maybe it's sort of gives a very harmonious hum uh and uh and sort of tries to give give Carriage the rest of the body so yeah I just uh there's an ecosystem there I'll go to another question in the live chat so Bert asks what do you think of the average of solving within a generative hierarchy and the impact of temporary openness moving it upwards and then I asked to clarify what do you mean the average of solving and Bert wrote where predictions meet incoming signals like discussed in Rebus where top down predictions are softened in precision and dust and thus solved for oh I think Arthur will be able to get into the details of that better than I would but the uh one thing I'm wondering there is um sometimes like like in like again I love the entropic brain uh but sometimes I wonder in terms of like the connections whether I don't so how we're having more or less entropy and whether that matters the how so for instance like you can think of like a free energy landscape that's been flattened or you can think of one where like you're better able and you have like more like energy for like this uh you know the manifold activity is better of like climbing up or down the the Peaks that are there for the more enduring belief landscape so it's like in the moment it could be that um uh my previous way of thinking is unavailable to me and I no longer believe those things at any level um or I I can uh support more novel inferences that I wouldn't otherwise I can kind of move into an area of Mind space that would otherwise be difficult potentially because of of greater energy into the system greater thorough put nice yeah so as I as I think I understand the question right it's basically asking about trying to optimize for this free energy variable and in the con the context of deep canals right like this is the loss function that's driving the optimization of the the learning landscape the synaptic great landscape right but like the entire connectivity structure of the nervous system is like trying to get to this point of like low-free energy and I think there's lots of uh there's you know for any given individual in a given environment assuming the environment is relatively fixed right there's like an optimal um there's a global optimal and it's

very likely that the individual is in some local Optimum that is maybe close to it maybe far from it but that's just inherently you know um what was you know within the context of the active entrance that's what we're assuming that the nervous system is doing is is doing this great into some in the space and openness then is going to uh correspond to this plasticity metatrade or be one of the components of it and it is in the Deep canals model then going to correspond to an ability to move around this optimization landscape more easily right it's going to be a flattening of a landscape or more precisely I guess a student of the um such that it's less likely that the individual the nervous system will get stuck in these local Optimum so maybe it won't mean that the individual will arrive at the global Optimum I think it's difficult to say what that even you know might be or however our individuals might get to such a place at all but I think it is likely that the individual will be will be more likely to find a more uh more Global Optimum and more easily move from one uh option to another right and I think that's kind of what this this openness is corresponding to it's okay oh I so in in neural kneeling I uh I mentioned the uh the idea of psychedelic extrapolating volition kind of a play on uh coherent extrapolated position from uh from the AI safety crowd and it's this sort of sense that um if temporarily you have more control over sort of low level shaping of of your your condition your emotion um and you know maybe sort of based on this this attractor that we call the South uh or you know I I guess like based on what is a good question um then what are good signs that you should be trusted with that power or you should trust yourself with that power rather and what are what are telltale signs that you shouldn't trust yourself with that power and um you know I I don't have an answer but I think it's an interesting question yes I am normal and can be trusted with the key to the doors of perception let me in exactly um I have a short question where is action in this model imagine many places Arthur will get into that but for um what I was thinking with um Albus is that actually the 5-HT_{2A} I I talk about two regimes potentially um one is more towards um overt in action and modifying the world and strength and beliefs potentially of that variety and the other might be more of um adapting to the world uh it's so uh Robin has a a really excellent paper called The Tale of Two receptors where he talks about 5-HT_{2C} two 5-HT_{1A} receptors the ones that antidepressants tend to act on um because they tend to reduce this excitatory signaling um talk about them as a kind of passive coping and so this would correspond and there's some machine learning models of this by Diane and colleagues of like where they talk about opponency between dopamine serotonin and they're focusing on the 1A as a kind of like patients promoting parameter keeping you from like confidently charging into your policy so that would tend to inhibit I guess patterns of

um either Overton action or potentially even some kinds of um uh mental simulation too if it's like you could maybe even think of like even if you're not overtly enacting something externally uh there might be more or less uh Brave type styles of thoughts that you might be willing to entertain going going places you might not otherwise um be willing to go and being able to handle the swings and expected free energy that might happen as you try to pursue a given path of policies so but the idea would be that um there might be something more like a micro dosing Mesa dosing regime which might be more of what initially sculpted the five hd2a system which evolved with the gene duplication event around the time of uh the Cambrian explosion with the Advent of jawed fishes I'm personally interpreting this as having the significance of um evolving the context of Predator prey arms races I suspect I could be wrong but that basically it's um locally trying to strengthen patterns of policy pursuit to pursue particular goals in a given context and this to come back to what was I think earlier without work by Robin of like a kind of active coping but it seems with Rebus um there's almost like um a not that kind of story um and that it's more your adjusting to the world and actually if under a very alter say you are overtly enacting that's actually kind of a dangerous thing I don't know how much to make of this at the um 582 receptors are less expressed on like motor cortex I don't think we know enough to make too much of it but it's like some dreams you might not want to act out as they're happening but they still might powerfully impact you so um I'm currently actually wondering whether this other Rebus like regime might have been specifically sculpted in the context of um bonding and actually opening the self to include others and that might have been some of the primary selective pressures I don't know how to think of that in terms of an action um but uh and one more detail that might kind of like uh not sure what to make of it the kind of goes through both of those excuse the phone uh would be um anthropologically I believe like psychedelics are oftentimes used in ways that would be both like creating a kind of group solidarity and might be used in um pursuing goals together in the world so for instance like a hunting party they might all do psychedelics together or something that looks a lot like breath work before uh going out maybe you know going to war so that's a very different type that kind of includes both a relaxation of the normal boundaries of the self and the kind of like uh gearing you up for pursuing goals in the world so but you know there's got to be another story of like like those Arthur's expertise is machine learning and so it's like uh these you want these systems to act intelligently and not act in other times and so basically now we're in the language of policies and when you deploy them and how and yes so complicated yeah I would say that action manifests itself in a few different places it's

trying to mention it here yeah I think it's really important obviously I mean I think what the kind of like most implicit place where it exists right is that um these belief Landscapes are being instantiated ultimately for the sake of adaptive Behavior adaptive Behavior adaptive action right in some sort of environmental context and that's the kind of like framework deep learning framework that we're using or like the artificial agent framework right is that like the assumption is that the system is supporting some sort of agent that's embodied in some sort of environment which is you know changing over time um such that it can either be adaptive or not adaptive I think then the question of how action actually plays into the evolution of these belief Landscapes you know during psychedelics or not you know acting in the world is a very clear way to modify one's setting right so action is a way of modifying the function that's providing the sensory input into the system which is going to evolve change how the activity patterns are evolving over time right which is going to affect the acute quality of the trip um it affects how beliefs are either being strengthened or relaxed um and I think you know it's it's already clear that there's a kind of whole that of like folk psychology right that's been built around like what is or is a good way to uh cultivate a helpful setting and that of course all is a function of action in the world I think that there's also um this kind of more subtler form of action which is how one thinks Holland deploys attention right essentially during this experience and I think this deployment of attention has a huge effect on whether there end up being Rebus or cbis like effects right so whether one allows oneself to go down certain thought rabbit holes whether one allows oneself to get kind of stuck in certain kinds of thought patterns I think this form of mental action yeah has a very big Downstream effect and I think a lot of I'd love to see much more work on this I think it's very important I think it's kind of essential um and I think this is kind of one of the most interesting things about these psychedelic experiences is that in some sense like the most extreme revised model and the most extreme uh Phoebe model could be true for two different individuals right someone could say you know I went to hell and I was there for eight hours straight or or say you know I had a completely selfless experience I like let go of all beliefs and experience you know nothing for eight hours straight it was great right and like um clearly those involve like two very different neural regimes and I think the nature and you know obviously given that they involve the same drug and they involve not such a different setting it's obvious that they um something very important to do with how attention is deployed during this experience yeah yeah just I do think that this Frame of attention I mean attention is so important for for everything and I do think that there's something interesting kind of digging into the the

attention and Elvis uh framework that um yeah I'm really curious to see how that'll that'll develop I I would also say that um in terms of active inference and action there is a cost to holding an attention um and then you once you know you discharge your intention you can release that cost um so I I guess I would have to think about how to apply that to the Canal's framework um Fred Barrett um I don't know if he's published it yet but this is kind of like a follow-up to a a cloud Sterling Central model that Arthur alluded to uh um is emphasizing basically um a lack of executive control and placing that is Central to the Psychedelic experience like it's not controlling your mental action and directed ways as much and then trying to come from more of the um types of cognition and phenomenology sort of psychedelics based on just releasing of uh the normal more agentic modes of policy selection around attention as mental action although he uh not using active inference speak so yes you have very interesting Adam you're your introduction there of of course attention as covert action which was echoed by by Arthur and Michael both of you said it really introduces uh a case where we can use active inference to find continuity between what's happening inside the agent and outside the agent because on one hand we could kind of take a belief-oriented frame and we could say well what's happening inside the agent are beliefs about things beliefs about hidden States beliefs about observations beliefs about policy so that's kind of a belief-oriented view where everything is framed it basically cognitively or we could take a very action-oriented View and then we could think about the interface in a more inactive way and then also even the realm of where where many active entities uh dare not tread in mental action with attention as covert action and so then we have like a a total active account that is mathematically isomorphic with the total belief-based accounts they're both just Maps they're not the territory the territories the person in the room so all of this is just talking about the formalism and how we choose to interpret the formalism in terms of it being more inferential or it being more active um and so there's a lot of ways to go I guess in our closing minutes I would love to hear what are the next moves or what are some areas that you um would hope that people look into Michael and then the authors okay so I know it's I think my my favorite mental move is to try to drop down on my level of abstraction okay how how is this implemented um what's what's going on here and then I do think that details of implementation um are really generative to look into and like to buy markers and proxies for you know figuring out how to have parameterize okay exactly how hot is are we talking about with this temperature parameter and so on and in different parts of the body um I I have kind of my my Niche that I'm I'm working on in the Vaso muscular system uh which is a rebital but

um yeah I just think that like I do think that this like the Deep canals framework specifically and the canalization Paradigm in general it's good enough that we should be able to dig into mechanisms so I'm I'm looking forward to that great Adam or Arthur the government doesn't so um or Arthur you can go first feel like it doesn't matter okay yeah okay sure um yeah I mean first I want to Echo this idea uh I think back to this is very important this is one of you know the people we've talked to about talked with about deep canals this idea of like well this is great you have these two axes you know it'd be wonderful if there was some like biological correlates you could look for um I think that's incredibly important um my background was also in machine learning uh in deep learning so I'm currently doing some work around like simulating some of these things I think uh especially for these Albus Dynamics being able to kind of like study under what circumstances you've got these different kinds of effects even in kind of like simple toy models is kind of important to see how they unfold and then I'm also you know we brought it up at the end I think it's really important it's very interesting to me is that the role of the deployment of attention in this um and there's lines of work where people are trying you know various researchers are trying to connect for example meditation to psychedelic research and kind of certain people are trying to tell unified stories about what's going on in the brain when people meditate what's going on they're just um some of them using these active imprints models um but I think in all of these cases the deployment of attention and how the deployment of tension changes over time is kind of a key aspect of this um and it's not as it's not as present in the Deep canals model right now but I any any really robust future kind of like model or these things is going to happen talk about attention quite a bit um I guess next steps would be um seeing how the review process goes but also I guess helping Arthur to establish a field of psychedelic machine learning psychedelic inspired machine learning where we use we try to make systems more flexible and capable better they can handle edge cases better imagine better in terms of real-time policy selection learn in ways that uh not too much not too little and then maybe you know so that's useful potentially earlier like you have a driverless vehicle do I change lanes or don't I you send it back to the Tesla Dojo what do you want to learn what do you not wanted to learn um but then down the line before we even get there you know as you're going into this you could see the capacity for like a rich bi-directional flow between machine learning and psychedelic inspired machine learning so it's like you get a bunch of machine Learners together and a bunch of people who know a lot about the cognitive science and biology and maybe phenomenology of psychedelics and in addition to designing better machine Learning Systems they could

probably tell you more about how psychedelics act and so you can have this kind of virtuous cycle between the biology and the field but you keep going with that um I guess before we get there we'll keep going but before we get there very interested in the implications for Best Practices so for instance like there might be like when so oftentimes you might want semantically neutral energy to kind of get them you're turning up the heat you're just maybe at first breaking up you know like home defenses but the usual structure but then the question is at what point do you try to direct the canalization process do you wait till after or maybe when the when the metal is hot is that a really good time to do it and so these are the kinds and maybe there's no one single question this might depend on the substance the set the setting the dose the person is so getting Clarity on that could be uh faithful for many people in terms of um the outcomes they have and I specifically what I'm uh gunning for is I'm on a mission to make uh compassion meditation uh treatment as usual for uh all psychedelic Psychotherapy and preferably All Therapy and so the question is like if you're in a state of enhanced imaginative capacity maybe you're having more ability to tune into your interceptive system whether you've just increased the channels or maybe you're better capable of because maybe your inflammation went down a bit you can tune in better because it doesn't feel it feels better to do so some combination that might be a really good opportunity to do certain meditations that might update you in particular ways like try to direct the re-annealing process maybe not just like afterwards integration but during the session and so if we could get clarity on something um how to catalyze and reanalyze in the like annealing schedules like it's going to be complicated but that's I think where we need to head and then finally um for all this I'm very uh that would be enough but um re-centering the conversation on psychedelics on basically meaning and connection would be enough but I'm wondering uh down the line when we talk of alignment we're talking about our alignment with our values and maybe the alignments of machines with our values does this actually end up going to some of the conversations that are happening now about increasingly Advanced Ai and our relationships to it could there be a story of canalization and reanalysis uh you want some things to hold fast and you want some things to be more fluid this might be it's always been a timely conversation we've always been dealing with this with us the alignment problem of us which we often fail but the alignment problem that these Technologies were developing wow well so many great topics thank you all I have in closing turn on tune in act in first serve thank you see you all next time